SOFTWARE ENGINNERING

LAB -5

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Sec:H

Before Alter the code:

scholarship_disbursement.py

```python
import os
import sqlite3

conn = sqlite3.connect(":memory:")
conn.execute("CREATE TABLE awards (id INTEGER PRIMARY KEY, student TEXT, scheme TEXT, amount INTEGER, disbursed INTEGER)")
disbursed_total = 0


def grant(student, scheme, amount, audit=[]):
    conn.execute("INSERT INTO awards (student, scheme, amount, disbursed) VALUES ('%s', '%s', %d, 0)" % (student, scheme, amount))
    audit.append(student)
    return audit


def disburse(award_id):
    global disbursed_total
    row = conn.execute("SELECT amount FROM awards WHERE id = ?", (award_id,)).fetchone()
    if row:
        conn.execute("UPDATE awards SET disbursed = 1 WHERE id = ?", (award_id,))
        disbursed_total += row[0]
        return row[0]


def archive():
    try:
        os.system("echo archiving awards")
    except Exception:
        pass


def bySchemeCount(scheme):
    if scheme == None:
        return 0
    return conn.execute("SELECT COUNT(*) FROM awards WHERE scheme = ?", (scheme,)).fetchone()[0]
```

```python
def main():
    grant("Ishaan", "Merit", 50000)
    grant("Divya", "Need-based", 75000)
    print(disburse(1), disburse(9))
    print(bySchemeCount("Merit"), disbursed_total)
    archive()


main()
```

Before Output of Python file:

```
PS C:\Users\shara> python -u "F:\scholarship_disbursement.py"
50000 None
1 50000
archiving awards
PS C:\Users\shara>
```

Before change pylint:

```
PS C:\Users\shara> python -m pylint "F:\scholarship_disbursement.py"
************* Module scholarship_disbursement
F:\scholarship_disbursement.py:5:0: C0301: Line too long (122/100) (line-too-long)
F:\scholarship_disbursement.py:10:0: C0301: Line too long (130/100) (line-too-long)
F:\scholarship_disbursement.py:1:0: C0114: Missing module docstring (missing-module-docstring)
F:\scholarship_disbursement.py:6:0: C0103: Constant name "disbursed_total" doesn't conform to UPPER_CASE naming style (invalid-name)
F:\scholarship_disbursement.py:9:0: C0116: Missing function or method docstring (missing-function-docstring)
F:\scholarship_disbursement.py:9:0: W0102: Dangerous default value [] as argument (dangerous-default-value)
F:\scholarship_disbursement.py:10:17: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
F:\scholarship_disbursement.py:15:0: C0116: Missing function or method docstring (missing-function-docstring)
F:\scholarship_disbursement.py:16:4: W0603: Using the global statement (global-statement)
F:\scholarship_disbursement.py:15:0: R1710: Either all return statements in a function should return an expression, or none of them should. (inconsistent-return-statements)
F:\scholarship_disbursement.py:24:0: C0116: Missing function or method docstring (missing-function-docstring)
F:\scholarship_disbursement.py:27:11: W0718: Catching too general exception Exception (broad-exception-caught)
F:\scholarship_disbursement.py:31:0: C0116: Missing function or method docstring (missing-function-docstring)
F:\scholarship_disbursement.py:31:0: C0103: Function name "bySchemeCount" doesn't conform to snake_case naming style (invalid-name)
F:\scholarship_disbursement.py:32:7: C0121: Comparison 'scheme == None' should be 'scheme is None' (singleton-comparison)
F:\scholarship_disbursement.py:37:0: C0116: Missing function or method docstring (missing-function-docstring)

-------------------------------------------------------------------
Your code has been rated at 5.00/10 (previous run: 5.00/10, +0.00)
```

Before change bandit

```
PS C:\Users\shara> python -m pylint "F:\scholarship_disbursement.py" > "F:\pylint_report.txt"
PS C:\Users\shara> python -m bandit "F:\scholarship_disbursement.py"
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.13.9
Run started:2026-10-14 06:35:35.442640+00:00

Test results:
>> Issue: [B608:hardcoded_sql_expressions] Possible SQL injection vector through string-based query construction.
   Severity: Medium   Confidence: Medium
   CWE: CWE-89 (https://cwe.mitre.org/data/definitions/89.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b608_hardcoded_sql_expressions.html
   Location: F:\scholarship_disbursement.py:10:17
9       def grant(student, scheme, amount, audit=[]):
10          conn.execute("INSERT INTO awards (student, scheme, amount, disbursed) VALUES ('%s', '%s', %d, 0)" % (student, scheme, amount))
11          audit.append(student)

--------------------------------------------------
>> Issue: [B605:start_process_with_a_shell] Starting a process with a shell: Seems safe, but may be changed in the future, consider rewriting without shell
   Severity: Low   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b605_start_process_with_a_shell.html
   Location: F:\scholarship_disbursement.py:26:8
25          try:
26              os.system("echo archiving awards")
27          except Exception:

--------------------------------------------------
>> Issue: [B607:start_process_with_partial_path] Starting a process with a partial executable path
   Severity: Low   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b607_start_process_with_partial_path.html
   Location: F:\scholarship_disbursement.py:26:8
25          try:
26              os.system("echo archiving awards")
27          except Exception:

--------------------------------------------------
```

```
--------------------------------------------------
>> Issue: [B607:start_process_with_partial_path] Starting a process with a partial executable path
   Severity: Low   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b607_start_process_with_partial_path.html
   Location: F:\scholarship_disbursement.py:26:8
25          try:
26              os.system("echo archiving awards")
27          except Exception:

--------------------------------------------------
>> Issue: [B110:try_except_pass] Try, Except, Pass detected.
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b110_try_except_pass.html
   Location: F:\scholarship_disbursement.py:27:4
26              os.system("echo archiving awards")
27          except Exception:
28              pass
29

--------------------------------------------------

Code scanned:
        Total lines of code: 32
        Total lines skipped (#nosec): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 3
                Medium: 1
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 1
                High: 3
Files skipped (0):
```

```
Files skipped (0):
PS C:\Users\shara> python -m bandit "F:\scholarship_disbursement.py" > "F:\bandit_report.txt"
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.13.9
PS C:\Users\shara>
```

Before change flake:

```
[main]  INFO    running on Python 3.13.9
PS C:\Users\shara> python -m flake8 "F:\scholarship_disbursement.py"
F:\scholarship_disbursement.py:5:80: E501 line too long (122 > 79 characters)
F:\scholarship_disbursement.py:10:80: E501 line too long (130 > 79 characters)
F:\scholarship_disbursement.py:17:80: E501 line too long (88 > 79 characters)
F:\scholarship_disbursement.py:19:80: E501 line too long (81 > 79 characters)
F:\scholarship_disbursement.py:32:15: E711 comparison to None should be 'if cond is None:'
F:\scholarship_disbursement.py:34:80: E501 line too long (96 > 79 characters)
PS C:\Users\shara>
```

After change the code:

```
PS C:\Users\shara> python -u "F:\scholarship_disbursement.py"
50000 None
1 50000
archiving awards
PS C:\Users\shara>
```

After change the Pylint:

```
PS C:\Users\shara> python -m pylint "F:\scholarship_disbursement.py"
************* Module scholarship_disbursement
F:\scholarship_disbursement.py:95:0: C0304: Final newline missing (missing-final-newline)

-------------------------------------------------------------------
Your code has been rated at 9.74/10 (previous run: 9.38/10, +0.36)

PS C:\Users\shara>
```

After change the bandit:

```
PS C:\Users\shara> python -m bandit "F:\scholarship_disbursement.py"
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.13.9
Run started:2026-10-14 07:01:06.692820+00:00

Test results:
        No issues identified.

Code scanned:
        Total lines of code: 76
        Total lines skipped (#nosec): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
Files skipped (0):
PS C:\Users\shara>
```

After the change flake:

```
PS C:\Users\shara> python -m flake8 "F:\scholarship_disbursement.py"
F:\scholarship_disbursement.py:95:11: W292 no newline at end of file
PS C:\Users\shara>
```